

Prioritized Episode Replay for RViT+ On-policy PPO with an off-policy auxiliary buffer

RViT+ implementation note

May 25, 2026

1 Motivation

The PPO loop in RViT+ collects a fresh batch of $B = \text{episodes_per_iter}$ on-policy episodes per iteration and discards them after the update. With a deterministic actor (e.g. “always wait”), this batch is dominated by a single (action, reward) class, depriving the contrastive auxiliary losses of the diverse pair structure they need. We introduce a prioritized episode replay buffer that augments each PPO update with a small number of *additional* episodes drawn from past iterations. The fresh on-policy batch is still used in its entirety; the replay batch is concatenated to it and weighted by importance-sampling (IS) coefficients so the prioritized expectation remains unbiased in the limit.

Why this is compatible with PPO. PPO’s clipped surrogate already absorbs mildly off-policy samples: a sample whose ratio $\pi_{\text{new}}/\pi_{\text{old}}$ moves outside $[1 - \epsilon, 1 + \epsilon]$ contributes zero gradient to the policy update. The distributional value loss, the reconstruction auxiliary, and the contrastive auxiliaries are all off-policy compatible. Therefore the only correction required is the standard PER IS weight.

2 Data Structure

The buffer is a FIFO ring of episodes, each represented as the contents of one row of a `RolloutBatch` (the observations $o_{0:T-1}$, executed actions $a_{0:T-1}$, rewards $r_{0:T-1}$, done flags, valid mask, behaviour log-probabilities $\log \pi_{\text{old}}(a_t|s_t)$, baseline values V_{old} , distributional baselines $V_{\text{old}}^{\text{dist}}$, bootstrap target $V_{\text{last}}^{\text{dist}}$, and episode length). Storage is on CPU; tensors move to the training device only when an episode is sampled. Each entry i has an associated scalar priority $p_i \geq 0$.

The buffer supports three operations:

- `push(batch, priorities)` — append every episode of `batch` with the supplied priorities (or with the current maximum priority if none are given).
- `sample(n,  $\alpha$ ,  $\beta$ )` — draw n episodes proportional to p_i^α and return them with their IS weights.
- `update_priorities(idxs, new_priorities)` — refresh the stored priorities after the sampled entries contribute to an update.

3 Priority Signal

Each episode’s priority is the mean per-step quantile-Huber error over its valid timesteps, evaluated under the current critic at the end of the PPO update. Let

$$\rho_{b,t}^{\text{QH}} = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N |\tau_i - \mathbf{1}(\delta_{ij} < 0)| \ell_{\kappa}(\delta_{ij}), \quad (1)$$

where $\delta_{ij} = y_j - q_i$ is the gap between target quantile y_j and predicted quantile q_i of the executed action at step (b, t) , $\tau_i = (i + 1/2)/N$ are the midpoint quantile fractions, and ℓ_{κ} is the Huber loss with parameter κ . Then the per-episode priority is

$$p_b = \frac{\sum_t m_{b,t} \rho_{b,t}^{\text{QH}}}{\sum_t m_{b,t}}, \quad (2)$$

with $m_{b,t} \in \{0, 1\}$ the valid mask. This is computed in `ppo_update` from the final epoch’s forward pass and returned to the trainer for the buffer refresh step.

4 Sampling Distribution

Given priorities $\{p_i\}_{i=1}^{N_{\text{buf}}}$, we first clip them at `per_priority_clip` · median(p) to prevent a single outlier episode from absorbing all sampling probability. We then form

$$P(i) = \frac{(\tilde{p}_i)^{\alpha}}{\sum_{j=1}^{N_{\text{buf}}} (\tilde{p}_j)^{\alpha}}, \quad \tilde{p}_i = \min(p_i, c \cdot \text{median}(p)), \quad (3)$$

where $\alpha \in [0, 1]$ controls how strongly priorities bias the distribution. $\alpha = 0$ recovers uniform sampling; $\alpha = 1$ recovers proportional priority sampling. We default to $\alpha = 0.6$ following Schaul et al. (2016).

5 Importance-Sampling Correction

Replacing uniform sampling with $P(i)$ biases any expectation over buffer entries. The standard PER correction multiplies each sample’s loss contribution by

$$w_i = \left(\frac{1}{N_{\text{buf}} \cdot P(i)} \right)^{\beta}, \quad \hat{w}_i = \frac{w_i}{\max_j w_j}, \quad (4)$$

where the max-normalisation keeps gradient magnitudes bounded across batches. Fresh on-policy episodes are not sampled from P , so they take $\hat{w}_i = 1$.

The exponent β governs how strongly we correct the bias. $\beta = 0$ is no correction (full prioritization bias); $\beta = 1$ is full correction (the expectation is exactly that of uniform sampling). We linearly anneal β from `per_beta_start` (default 0.4) to `per_beta_end` (default 1.0) across training:

$$\beta(t) = \beta_{\text{start}} + (\beta_{\text{end}} - \beta_{\text{start}}) \min\left(\frac{t}{T_{\text{train}} - 1}, 1\right). \quad (5)$$

Early in training, β is small to amplify the learning signal from prioritized samples; by the end of training, β reaches 1 and the prioritized expectation is unbiased.

6 Combined Batch and Weighted Losses

At each iteration t , the trainer constructs a combined batch by concatenating along the batch dimension:

$$\mathcal{B}_t = \mathcal{B}_t^{\text{fresh}} \parallel \mathcal{B}_t^{\text{replay}}, \quad B_{\text{tot}} = B + B_{\text{replay}}, \quad (6)$$

where $\mathcal{B}_t^{\text{fresh}}$ contains the freshly collected B on-policy episodes (each with $\hat{w} = 1$) and $\mathcal{B}_t^{\text{replay}}$ contains $B_{\text{replay}} = \text{per_n_replay}$ episodes drawn from the buffer with their PER weights.

Let $\hat{w}_b$ denote the per-episode IS weight ($\hat{w}_b = 1$ for fresh, $\hat{w}_b$ given by (4) for replay), and $m_{b,t}$ the valid-step mask. The *effective weight* is

$$\mu_{b,t} = m_{b,t} \cdot \hat{w}_b, \quad M = \sum_{b,t} \mu_{b,t}. \quad (7)$$

Every loss term that took a masked mean now takes a weighted masked mean using μ :

$$\mathcal{L}_{\text{policy}} = -\frac{1}{M} \sum_{b,t} \mu_{b,t} \min(r_{b,t} A_{b,t}, \text{clip}(r_{b,t}, 1-\epsilon, 1+\epsilon) A_{b,t}), \quad (8)$$

$$\mathcal{L}_{\text{value}} = \frac{1}{M} \sum_{b,t} \mu_{b,t} \rho_{b,t}^{\text{QH}}, \quad (9)$$

$$\mathcal{L}_{\text{entropy}} = -\frac{1}{M} \sum_{b,t} \mu_{b,t} \mathcal{H}[\pi(\cdot | s_{b,t})], \quad (10)$$

where $r_{b,t} = \pi_{\theta}(a_{b,t} | s_{b,t}) / \pi_{\text{old}}(a_{b,t} | s_{b,t})$ is the PPO ratio and $A_{b,t}$ are the GAE advantages computed *at collection time* for both fresh and replay entries. The reconstruction auxiliary uses the same content-weighted mask scaled by $\hat{w}_b$. The contrastive auxiliaries operate on pairs of valid (b, t) entries; their pair masks are formed without further IS correction because the contrastive losses shape representations rather than estimate an unbiased expectation.

7 Algorithm

Algorithm 1 PPO update with prioritized episode replay

```

1: Initialise ring buffer  $\mathcal{R}$  of capacity  $C$ ; optimizer  $\mathcal{O}$ .
2: for  $t = 0, 1, \dots, T_{\text{train}} - 1$  do
3:    $\mathcal{B}_t^{\text{fresh}} \leftarrow$  collect  $B$  on-policy episodes
4:    $\beta_t \leftarrow \beta_{\text{start}} + (\beta_{\text{end}} - \beta_{\text{start}}) \min(t/(T_{\text{train}} - 1), 1)$ 
5:   if  $|\mathcal{R}| > 0$  and  $B_{\text{replay}} > 0$  then
6:     Compute  $P(i)$  per Eq. (3)
7:     Sample indices  $I = \{i_1, \dots, i_{B_{\text{replay}}}\} \sim P$ 
8:      $\hat{w}_k \leftarrow (1/(N_{\text{buf}} \cdot P(i_k)))^{\beta_t}$ , then  $\hat{w}_k \leftarrow \hat{w}_k / \max \hat{w}$ 
9:      $\mathcal{B}_t^{\text{replay}} \leftarrow \mathcal{R}[I]$  with weights  $\hat{w}$ 
10:  else
11:     $\mathcal{B}_t^{\text{replay}} \leftarrow \emptyset, I \leftarrow \emptyset$ 
12:  end if
13:   $\mathcal{B}_t \leftarrow \mathcal{B}_t^{\text{fresh}} \parallel \mathcal{B}_t^{\text{replay}}$  ▷ Eq. 4
14:  for  $k = 1, \dots, n_{\text{epochs}}$  do
15:    Forward + compute  $\rho_{b,t}^{\text{QH}}$  per (2)
16:    Compute weighted losses  $\mathcal{L}_{\text{policy}}, \mathcal{L}_{\text{value}}, \dots$ 
17:     $\mathcal{O}.\text{step}(\nabla_{\theta} \mathcal{L})$ 
18:  end for
19:  Compute final per-episode priorities  $\{p_b\}_{b=1}^{B_{\text{tot}}}$  per Eq. (2)
20:   $\mathcal{R}.\text{update\_priorities}(I, \{p_b\}_{b>B})$  ▷ refresh replay entries
21:   $\mathcal{R}.\text{push}(\mathcal{B}_t^{\text{fresh}}, \{p_b\}_{b \leq B})$  ▷ enqueue fresh
22: end for

```

8 Bias Analysis

The Monte-Carlo estimator of any per-transition expectation $\mathbb{E}_{i \sim \text{uniform}}[f(i)]$ under prioritized sampling is

$$\hat{\mathbb{E}}[f]_{\text{PER}} = \frac{1}{n} \sum_{k=1}^n \frac{1}{N_{\text{buf}} P(i_k)} f(i_k), \quad i_k \sim P. \quad (11)$$

With the full IS weight $w_i = 1/(N_{\text{buf}} P(i))$ (i.e. $\beta = 1$), $\mathbb{E}_P[\hat{\mathbb{E}}] = \mathbb{E}_{\text{uniform}}[f]$ exactly. With $\beta < 1$ the estimator is biased toward the prioritized distribution. The max-normalisation $\hat{w} = w / \max w$ rescales the gradient magnitude but does not change the relative weighting; it preserves the unbiased property in the limit $\beta \rightarrow 1$ up to an overall constant.

The priority clipping in Eq. (3) introduces a small controlled bias even at $\beta = 1$, because the truncated distribution $\tilde{p}$ differs from the true p at extreme entries. This is intentional — we trade a small expectation bias for protection against a single high-error episode capturing all sampling probability.

9 Hyperparameters and Defaults

Symbol	Config key	Default
C (capacity)	<code>buffer_capacity</code>	200 episodes
B_{replay}	<code>per_n_replay</code>	4
α	<code>per_alpha</code>	0.6
β_{start}	<code>per_beta_start</code>	0.4
β_{end}	<code>per_beta_end</code>	1.0
c (priority clip mult.)	<code>per_priority_clip</code>	50.0

Setting `per_n_replay` = 0 disables replay and restores pure on-policy PPO.

10 Empirical Observation (Confirmed, 2026-05-25)

Status upgrade. The result documented below was originally recorded as preliminary based on short-horizon training. As of 2026-05-25 it is *confirmed* as the first sustained task-competent RViT+ configuration on Posner change-detection. At iter $\approx 1450\text{--}1670$ (over 200 iterations of steady-state behaviour) the model holds `correct` $\in [0.70, 0.78]$, `return` $\in [2.1, 2.3]$, episode `length` $\in [22, 28]$ out of $T = 29$, with policy entropy $H[\pi] \in [0.01, 0.04]$ — committed without being deterministic-collapsed. The full working baseline combines PER (this document) with predictive-coding auxiliaries, contrastive auxiliaries, and the PAC actor loss (Springenberg et al., 2024). See `research_db/threads/rvit_plus_engineering.md` section “Working baseline (2026-05-25)” for the complete loss decomposition and hyperparameter table.

Mechanism catalogue (preserved from preliminary notes). Across the RViT+ development arc — adaptive certainty-decay σ -gates, intrinsic UCB exploration via critic uncertainty, actor-pull terms weighted by Q-entropy, per-action critic-decay penalties on unchosen actions, single-cell and triplet-input contrastive heads with binary and continuous-weight loss formulations, multiple coefficient sweeps, and several margin / scale schedules — *prioritized episode replay is the first mechanism that empirically breaks the symmetry of the encoder’s internal representations and holds the actor’s output entropy above the deterministic-collapse floor*. Every earlier approach either had no discernible effect, collapsed the actor to a single action within $\mathcal{O}(10^2)$ iterations, or pinned the auxiliary gate σ to its boundary.

This result is not yet conclusive. The current findings are based on short-horizon training runs and the long-term dynamics under PER on this architecture remain to be characterised. We are not claiming that PER will succeed where the symmetry-shaping losses failed; we are only recording that PER is, at this stage, the only mechanism that has produced the qualitative signature we were looking for from the others: persistent representation diversity coupled to a non-collapsed policy. We consider this sufficiently promising to make PER the default training configuration going forward.

Gradient-magnitude restoration. A companion empirical observation: the max-absolute gradient element `gmax` jumps by roughly four-to-five orders of magnitude between the pre-PER and post-PER regimes. Pre-PER (any of the symmetry-shaping mechanisms above, in their collapsed-actor regimes) `gmax` settles around 10^{-2} . Post-PER, after the buffer warms and replay episodes begin contributing, `gmax` settles in the $10^2\text{--}10^3$ range. The two-stage gradient clip (per-element cap at 10^6 and L2-norm cap at 0.5) handles this magnitude cleanly and the individual loss terms

remain bounded, so the change is not a stability concern. Rather, the collapsed batch was producing essentially no contrast signal: all pairs shared the same (a, r) class, the value loss had no novel transitions to fit, and the per-step quantile-Huber error was uniformly small. PER reintroduces diverse, informative transitions into every update; the resulting gradient magnitude is the characteristic regime of an actively-learning network rather than the flat-collapse-point regime. The restoration tracks the contrastive cosine separation 1:1 across iterations.

11 Practical Notes

Pretraining. During the reconstruction pretraining phase ($t < \text{recon_pretrain_iters}$) we still push fresh episodes into the buffer (with initial priority equal to the running maximum), but do not sample from it. This warms the buffer before PPO begins.

Stale-policy gradient. A replay sample collected k iterations ago has a behaviour policy π_{old} that may have drifted from π_{θ} . PPO’s clip range ϵ silently zeroes the policy gradient whenever $r_{b,t}$ moves outside $[1 - \epsilon, 1 + \epsilon]$, which means stale replay samples contribute essentially nothing to $\mathcal{L}_{\text{policy}}$. The value, reconstruction, and contrastive losses remain off-policy compatible and continue to benefit from the additional data.

Mixing ratio. We keep $B_{\text{replay}} \leq B$ in practice (default 4 vs 8) so that on-policy episodes always dominate the combined batch. This preserves PPO’s locally-optimal step direction while still injecting the rare-event coverage the auxiliary losses need.

Logging. Each iteration the trainer prints buffer fill, current β , and the number of replay episodes used:

`buf=8/200    $\beta$ =0.42   +rep=4`

which can be cross-referenced with the contrastive cosine diagnostics `cos[+/-]` to confirm that replay is in fact increasing pair diversity.